Appendix1. Description of Information System

Application No.: (To be filled in by Contest Committee)
Title of Information System: Haru: A Daily Cognitive Routine Kiosk with Speech Recognition and Memory-Grounded Personalization
University / Institution: Institute for Universal Education
Department / Faculty: Haru Development Team

I. Preface

Haru is an institution-installed daily cognitive routine for adults aged 65 and above. It addresses the gap between occasional phone calls and repeated care observation: older adults may struggle with small smartphone screens, while institutions cannot collect the same structured record every day. Haru offers a calm three-to-five-minute routine that supports conversation and professional follow-up; it is not a diagnostic device.

II. Innovation Description

Haru combines an accessible 2×2 physical-button kiosk, speech-to-text recording, and memory-grounded question personalization. Its innovation is a closed loop: an easy daily interaction becomes a structured personal record and then a relevant next-day question.

Accessible routine → **Voice / button record** → **Related memory retrieval** → **Next-day prompt** → **Support report**

Screen positions and button positions correspond. The first press selects an answer and the same button pressed again confirms it, so participants do not need to memorize button labels or learn a separate cancel flow.



Physical prototype



Button interaction

Figure 1. Team prototype photographs reproduced from the E5 presentation materials.

III. System Functions

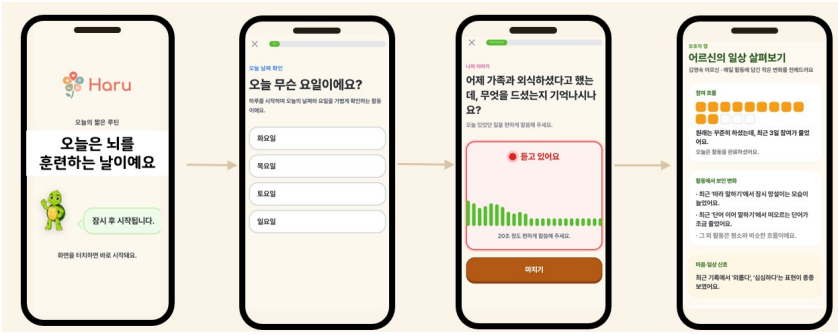
Participant access. An NFC tap opens the routine with a pseudonymous card ID; the device does not display the participant's name.

Routine and speech. Six short activities run once per day across a 42-activity, seven-day

structure. Voice prompts and recordings are transcribed by STT; a retry path keeps recognition failure from blocking completion.

Personalized prompt. Past responses are stored with dates and extracted cues. Related memories are retrieved to ground a follow-up question instead of selecting one at random.

Role-based report. Participants see their routine result; caregivers see consented conversation cues; welfare staff manage registration and participation; counselors or clinicians review summarized observations.



Routine-to-report screen sequence

Figure 2. User-facing screens reproduced from the E5 presentation materials.

IV. System Features

Accessibility and recovery. Large type, high contrast, four large touch targets, spoken guidance, and spatially matched buttons reduce visual and menu burden. Separate selection and confirmation make mistakes recoverable; speech can be retried without cancelling the session.

Grounded personalization. Each generated prompt is constrained by a retrieved personal-memory record with a date and topic, so the source of the prompt can be explained rather than hidden behind a random question.

Privacy and safety. Names stay with the registering institution. The kiosk uses a pseudonymous ID, consent scopes, role-based access, and deletion or withdrawal procedures. Reports describe supportive changes, not a diagnosis or clinical score.

Validation readiness. A formative 20-person beta test drove the kiosk redesign. A separate IRB study will test independent target-age participants in Korea and Japan.

V. System Development Tools and Techniques

The system uses a browser kiosk client and a speech-and-memory pipeline. The client presents the routine and records events; the services return structured evidence for reports and future prompts.

Audio capture → **STT** → **Cue extraction** → **Memory index** → **RAG prompt**

Speech recognition. Older speech includes noise, overlap, distance, slow rate, long pauses, and low volume. On FLEURS, Qwen3-ASR versus Whisper large-v3 achieved Korean CER 2.96% versus 3.01% and Japanese CER 5.28% versus 5.56%. The selected engine is reused for activity records and personalization.



Existing STT architecture figure

Technical figure specification — STT pipeline (redraw from the existing infographic)

Keep the microphone, waveform, and comparison motifs, but redraw a left-to-right pipeline: microphone audio → pause/noise-aware capture → selected ASR engine → transcript and CER → activity record. Place the Korean and Japanese FLEURS values beside the model block. Use "team-selected engine" unless fine-tuning is verified.

Technical figure specification — RAG personalization (extend the existing conceptual flow)

Extend the existing story-record → memory-store → related-memory-search → new-question flow with the missing layer: extract date, person, place, event, emotion, and consent fields; embed the memory; retrieve top related records; pass only those records and a template to the language model; write back the prompt with source-memory IDs. Add a small example: "I ate out with my daughter yesterday" → "What did you have when you ate out with your daughter?" Use separate colors for personal data and generated text.

VI. System Users

The system separates the learner-facing experience from the support-facing views. Access is scoped to the role and to the consent recorded by the registering institution.

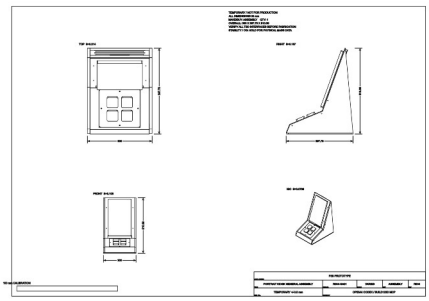
User	Primary use
Older adult	Taps an NFC card, completes six activities, and answers by voice or button.

Welfare-center staff	Registers consent and the pseudonymous card ID, then checks participation.
Caregiver	Views consented check-in and conversation cues, not a clinical score.
Counselor / clinician	Reviews summarized observations to prepare a professional conversation.

The 20-person beta test is formative usability evidence. The IRB study will test independent participants, consent, speech performance, completion burden, and report usefulness.

VII. System Environment

Layer	Description
Hardware	Raspberry Pi, portrait display, four 2×2 buttons, NFC reader, microphone, speaker, and enclosure.
Client software	A large-text kiosk web client for one-person, click-first operation; Korean and Japanese are supported.
Data path	The card supplies a pseudonymous ID. Responses, transcripts, and consent-scoped summaries stay separate from names.
Operating site	Staff register cards and participants; the older adult starts by tapping the card at the installed kiosk.



Prototype fabrication drawing

Figure 3. Prototype fabrication drawing reproduced from the E5 presentation materials.

VIII. Conclusion

Haru closes a practical care loop: a low-friction physical interaction makes daily participation possible; STT turns spoken answers into searchable records; RAG uses those records to create relevant next-day prompts; and consent-scoped summaries give families and care professionals material for a conversation. The prototype, NFC flow, bilingual interface, report path, and formative usability changes are implemented.

The next step is field validation, not a clinical claim: the team will complete the IRB process, run the Korea and Japan studies, measure independent completion, time, speech-recognition success, repeat-use intention, and report usefulness, and then refine the product. Any final technical figure should be checked against the running data pipeline, especially the exact embedding/index implementation and the verified status of the selected ASR model.

Submission note

All photographs, screenshots, diagrams, and prototype drawings in this description are team-owned assets reproduced from the supplied Haru presentation materials. The system is presented as a supportive routine and observation aid, not as a substitute for professional diagnosis.